作业4. 思考题

1、step1：设过原点的平面方程为

$\Pi: x\cos\alpha + y\sin\alpha = \mu z$，$\mu \neq 0$.

由于其具有良好的旋转对称性系统，故可取$\alpha=0$时证明

$\therefore \Pi: x = \mu z$.

step2：引入辅助平面 $Q: y=0$.

$\therefore$ Q与Π交线为 $L: x=\mu z$，$y=0$.

Q与H交线为(z支)：$x=\sqrt{z^2-1}$

$\therefore$两交线的交点 P：满足 $\mu z = \sqrt{z^2-1} \Rightarrow z = \frac{1}{\sqrt{1-\mu^2}}$

$\therefore P\left(\frac{\mu}{\sqrt{1-\mu^2}}, 0, \frac{1}{\sqrt{1-\mu^2}}\right)$

step3：引入柱坐标 $x=r\cos\theta$，$y=r\sin\theta$，$z=z$.

$\therefore H: z^2-r^2=1$ 即 $r=\sqrt{z^2-1}$

$\Pi: r\cos\theta = \mu z$

$\therefore$渐线 γ：$\begin{cases} r=\sqrt{z^2-1} \\ \cos\theta = \frac{\mu z}{\sqrt{z^2-1}} \end{cases}$

step4：以φ为从顶点$(0,0,-1)$指向γ上等定Q与z轴的夹角

$\therefore \tan\varphi = \frac{r}{z+1} = \frac{\sqrt{z^2-1}}{z+1} = \sqrt{\frac{z-1}{z+1}}$

$\therefore \tan\varphi_p = \frac{\sqrt{z_p^2-1}}{z_p+1} = \sqrt{\frac{z_p-1}{z_p+1}}$

$\therefore z = \frac{1}{\cos 2\varphi}$；$r=\sqrt{z^2-1} = \tan 2\varphi$

代入 $r\cos\theta = \mu z$.

$\therefore \tan 2\varphi \cos\theta = \mu \cdot \frac{1}{\cos 2\varphi}$

$\sin 2\varphi \cos\theta = \mu$

step5: 令顶点 $V(0,0,-1)$, VQ 交单位球面于 Q'

$\therefore Q'(\sin\varphi\cos\theta, \sin\varphi\sin\theta, \cos\varphi)$, 记为 (x, y, z)

$\because 2\sin\varphi\cos\varphi\cos\theta = \mu$

$\therefore 2xz = \mu \Rightarrow x = \frac{\mu}{2z}$

设 $Q(\mu z, y, z)$. $\therefore y^2 = z^2(1-\mu^2) - 1$

$\therefore VQ: L(t) = V + t(Q - V)$

$= (t\mu z, ty, -1 + t(z+1))$, $t \in R$

$\therefore Q': (t\mu z)^2 + (ty)^2 + (-1 + t(z+1))^2 = 1$

$t^2(z^2 - 1) + (-1 + t(z+1))^2 = 1$

$2z(z+1)t^2 - 2t(z+1) = 0$ $z \neq -1$

$2zt^2 - 2t = 0$ $\therefore t = \frac{1}{z}$

将 $t = \frac{1}{z}$ 代入 $L(t)$

$\therefore Q' = (\mu, \frac{y}{z}, \frac{1}{z})$, Q' 在 $x = \mu$ 这个竖直平面上

令 $Y = \frac{y}{z}$, $Z = \frac{1}{z}$

$z^2 - (\mu z)^2 - y^2 = 1$ $\therefore (\frac{y}{z})^2 + \frac{1}{z^2} = 1 - \mu^2$

即: $Y^2 + Z^2 = 1 - \mu^2$ $\therefore Q'$ 在一个圆弧上, 圆心在平面 $z = 0$ 上

$\therefore$ 交线 C 是竖直平面内圆弧且圆心在 $z = 0$ 上

2. 从V到Q'的直线参数方程为

$$l(t)=(0,0,-1)+t(\sin\varphi\cos\theta,\ \sin\varphi\sin\theta,\ \cos\varphi+1)$$

令 $z=0$ $\quad\therefore t=\dfrac{1}{\cos\varphi+1}$

$\therefore$ 交点 P: $\left(\dfrac{\sin\varphi\cos\theta}{\cos\varphi+1},\ \dfrac{\sin\varphi\sin\theta}{\cos\varphi+1},\ 0\right)$.

取极坐标 $x=r\cos\theta$, $y=r\sin\theta$.

$$\therefore r=\frac{\sin\varphi}{\cos\varphi+1}=\tan\frac{\varphi}{2}$$

又 $\quad\because z=\dfrac{r\cos\theta}{\mu}$

$$\therefore\left(\frac{r\cos\theta}{\mu}\right)^2-r^2=1.$$

$$r^2\left(\frac{\cos^2\theta}{\mu^2}-1\right)=1 \qquad \therefore r=\frac{\mu}{\sqrt{\cos^2\theta-\mu^2}}$$

$$z=\frac{\cos\theta}{\sqrt{\cos^2\theta-\mu^2}} \qquad \therefore x=r\cos\theta=\frac{\mu\cos\theta}{\sqrt{\cos^2\theta-\mu^2}}$$

$$\therefore\tan\varphi=\frac{x}{\dfrac{\cos\theta}{\sqrt{\cos^2\theta-\mu^2}}+1}=\frac{\mu}{\cos\theta+\sqrt{\cos^2\theta-\mu^2}}$$

$$\therefore\varphi=\arctan\frac{\mu}{\cos\theta+\sqrt{\cos^2\theta-\mu^2}},$$

$$r=\tan\frac{1}{2}\arctan\frac{\mu}{\cos\theta+\sqrt{\cos^2\theta-\mu^2}} \qquad (*)$$

记 $r=\tan\dfrac{\phi}{2}=\dfrac{\sin\phi}{1+\cos\phi}$

设直角坐标下圆方程：$(x-a)^2+y^2=R^2$.

$\therefore$ 极坐标：$r^2-2a\cos\theta r+(a^2-R^2)=0$. ①

令 $\theta=0$ $\therefore \tan\varphi_0=\dfrac{\mu}{1+\sqrt{1-\mu^2}}$, $r_0=\tan\dfrac{\varphi_0}{2}$.

令 $\theta=\arccos\mu$ $\therefore \tan\varphi_1=1$, $\varphi_1=\dfrac{\pi}{4}$, $r_1=\tan\dfrac{\pi}{8}$.

令 $\theta=\pi$ $\therefore \tan\varphi_\pi'=\dfrac{\mu}{-1+\sqrt{1-\mu^2}}$, 取 $\varphi_\pi=\dfrac{\mu}{1-\sqrt{1-\mu^2}}$

$\therefore r_\pi=\tan\dfrac{\varphi_\pi}{2}$

将点 r_0 和点 r_π 代入

$$\therefore \begin{cases} r_0^2-2ar_0+(a^2-R^2)=0. \\ r_\pi^2-2ar_\pi+(a^2-R^2)=0. \end{cases} \Rightarrow \begin{cases} a=\dfrac{r_0-r_\pi}{2} \\ R=\dfrac{r_0+r_\pi}{2} \end{cases}$$

$\because r^2-2a\cos\theta r+(a^2-R^2)=0$.

$\therefore r=a\cos\theta\pm\sqrt{R^2-a^2\sin^2\theta}$. 将 a 与 R 代入

设 $A=r_0$, $B=r_\pi$.

$\therefore r=\dfrac{A-B}{2}\cos\theta\pm\dfrac{1}{2}\sqrt{(A+B)^2-(A-B)^2\sin^2\theta}$.

$=\dfrac{A-B}{2}\cos\theta\pm\dfrac{1}{2}\sqrt{(A^2+B^2)\cos^2\theta+2AB(1+\sin^2\theta)}$ ②.

$\because AB=1$, $A^2+B^2=(A+B)^2-2$.

$\therefore (A^2+B^2)\cos^2\theta+2AB(1+\sin^2\theta)$

$=((A+B)^2-2)\cos^2\theta+2+2\sin^2\theta=(A+B)^2\cos^2\theta+2-2\cos2\theta$. ③

$\because A+B=2\sec\left(\arctan\dfrac{\mu}{\cos\theta+\sqrt{\cos^2\theta-\mu^2}}\right)$ ④

$A-B=2\tan\left(\arctan\dfrac{\mu}{\cos\theta+\sqrt{\cos^2\theta-\mu^2}}\right)$ ⑤ 代入.

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将③④⑤代入②.

$$r=\tan\left(\frac{1}{2}\arctan\frac{\mu}{\cos\theta+\sqrt{\cos^2\theta-\mu^2}}\right),$$

与步骤1中方程相同.

故：交线C'也是圆弧

下证：此圆与单位圆正交

单位圆在$\alpha=0$平面方程为 $x^2+y^2=1$

$\because a=\frac{r_0-r_\pi}{2}$, $R=\frac{r_0+r_\pi}{2}$ $\therefore$ 由前有 $r_0r_\pi=1$

$\therefore R^2=\frac{1}{4}(r_0^2+r_\pi^2+2r_0r_\pi)$

$a^2=\frac{1}{4}(r_0^2+r_\pi^2-2r_0r_\pi)$

$=R^2-1$ 即：$a^2+1=R^2$

$\therefore$ C'与单位圆垂直相交.

即：C'也是圆弧，且与单位圆垂直相交.